

Paired-Difference Tests II Use the information in Exercise 9 to calculate the observed value of the t statistic for testing the difference between the two population means using paired data. Find a rejection region with $\alpha = .05$ to state your conclusions. Construct a 95% confidence interval for $\mu_1 - \mu_2$. Does this confirm the results of your hypothesis test?

9.

Population	Pairs				
	1	2	3	4	5
1	1.3	1.6	1.1	1.4	1.7
2	1.2	1.5	1.1	1.2	1.8
d	0.1	0.1	0	0.2	-0.1

$$\bar{d} = 0.06 \quad S_d^2 = 0.013 \quad S_d = 0.114$$

$$H_0: \mu_d = 0 \quad H_a: \mu_d \neq 0$$

$$t^* = \frac{0.06 - 0}{\frac{0.114}{\sqrt{5}}} \approx 1.176$$

$$t_{4; 0.025} = 2.776$$

rejection region: $t > |2.776| \Rightarrow$ Not reject

$$\begin{aligned} CI: \bar{d} \pm t_{4; 0.025} \cdot \frac{S_d}{\sqrt{n}} \\ = 0.06 \pm 2.776 \cdot 0.051 \\ = (-0.0816, 0.2016) \end{aligned}$$

Conclusion:

Yes, CI include 0.

Which means difference = 0 is plausible

DATA
SET

DS1026

10. Memory Experiments Twenty students participated in an experiment in which they were first asked to recall a list of 25 words. They were then given a second list, and asked to form images of the words as they were read. The number of words recalled for each student is given in the table.

Student With Imagery Without Imagery

$$\bar{D} = 10.25$$

$$S_D = 4.051$$

$$H_0: D_0 = 0, \quad H_a: D_0 > 0, \quad \alpha = 0.05$$